

AEROSPACE ENGINEERING [AERODYNAMICS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.620
Fluid Dynamics I

Summer/Fall

535.626
Mechanics of Flight

Fall/Spring

535.670
Advanced Aerodynamics

Fall

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.625
Turbulence

Fall

535.626
Mechanics of Flight

Fall/Spring

535.639
Aerospace Materials

Spring

535.652
Thermal Systems Design and
Analysis

Fall

535.670
Advanced Aerodynamics

Fall

535.735
Computational Fluid Dynamics

Fall

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.761
Hypersonic Aerothermodynamics

Spring

535.763
Aerospace Propulsion

Spring

AEROSPACE ENGINEERING [DYNAMICS & CONTROL FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.626
Mechanics of Flight

Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

Group 2 Courses

535.608
Hypersonic Technologies and
Systems

Fall

535.612
Intermediate Dynamics

Fall

535.623
Intermediate Vibrations

Fall/Spring

535.626
Mechanics of Flight

Fall/Spring

535.642
Control Systems for Mechanical
Engineering Applications

Fall

535.670
Advanced Aerodynamics

Fall

535.724
Dynamics of Robots and
Spacecraft

Spring

535.741
Optimal Control and
Reinforcement Learning

Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles

Fall

535.762
Guidance, Navigation and
Controls for Hypersonic Vehicles

Fall/Spring

535.763
Aerospace Propulsion

Spring

525.609
Continuous Control Systems

Fall/Spring

525.645
Modern Navigation Systems

Fall/Spring

525.661
UAV Systems and Controls

Fall/Spring

525.777
Control System Design Methods

Fall

AEROSPACE ENGINEERING [STRUCTURES & MATERIALS FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.632
Applied Finite Elements
Every Term

Group 2 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.608
Hypersonic Technologies and
Systems
Fall

535.612
Intermediate Dynamics
Fall

535.623
Intermediate Vibrations
Fall/Spring

535.627
Computer-Aided Design
Every Term

535.628
Computer-Integrated Design &
Manufacturing
Every Term

535.632
Applied Finite Elements
Every Term

535.639
Aerospace Materials
Spring

535.643
Plasticity
Spring

535.652
Thermal Systems Design and
Analysis
Fall

535.660
Precision Mechanical Design
Fall/Spring

535.684
Modern Polymeric Materials
Fall/Spring

535.706
Mechanics of Solids & Structures:
Theory and Applications II
Spring

535.720
Mechanics of Composite
Materials and Structures
Fall

535.721
Advanced Composite Materials &
Manufacturing Processes
Spring/Summer

535.731
Engineering Materials: Properties
and Selections
Fall/Spring

535.732
Fatigue and Fracture of Materials
Spring

515.658
Design for Additive Manufacturing
Fall

515.661
Introduction to Polymer Science
Fall

675.702
Spacecraft Materials
Every Term

675.722
Space Mechanical Design and
Analysis
In Development

AEROSPACE ENGINEERING [VEHICLE DESIGN FOCUS AREA]

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.608
Hypersonic Technologies and
Systems
Fall

535.626
Mechanics of Flight
Fall/Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles
Fall

Group 2 Courses

535.606
Advanced Strength of Materials
Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I
Fall

535.608
Hypersonic Technologies and
Systems
Fall

535.626
Mechanics of Flight
Fall/Spring

535.627
Computer-Aided Design
Every Term

535.628
Computer-Integrated Design &
Manufacturing
Every Term

535.632
Applied Finite Elements
Every Term

535.652
Thermal Systems Design and
Analysis
Fall

535.660
Precision Mechanical Design
Fall/Spring

535.670
Advanced Aerodynamics
Fall

535.706
Mechanics of Solids & Structures:
Theory and Applications II
Spring

535.720
Mechanics of Composite
Materials and Structures
Fall

535.721
Advanced Composite Materials &
Manufacturing Processes
Spring/Summer

535.724
Dynamics of Robots and
Spacecrafts
Spring

535.752
Advanced Flight Dynamics and
Control of Aerospace Vehicles
Fall

535.761
Hypersonic Aerothermodynamics
Spring

535.763
Aerospace Propulsion
Spring

675.722
Space Mechanical Design and
Analysis
In Development